ECE 562: Deep Reinforcement Learning for Robotics

Assignment 2: Proximal Policy Optimization (PPO)

Emmanuel Olateju
eolateju Student ID: 1984858

Fall 2026

Contents

1 Theoretical Foundations of PPO **2**

2 Continuous Control Implementation in MuJoCo **14**
2.1 Q6(a) — Benchmark: Default Hyperparameters (3 Seeds) 14
2.2 Q6(b) — GAE λ Trade-off Ablation 15
2.3 Q6(c) — Policy Clipping Ablation ($\epsilon = \infty$) 16

1 Theoretical Foundations of PPO

The following pages contain the handwritten theoretical derivations for Section 1 (Questions 1 and 2), covering Generalized Advantage Estimation and the clipped surrogate objective as a pessimistic lower bound.

EMMANUEL OLUWATOSIN OLATEJU

DEEP RL FOR ROBOTICS CEID: eolateju

ASSIGNMENT TWO STUDENT ID : 1984858

9.3 THEORETICAL FOUNDATIONS OF PPO

Q1] Generalized Advantage Estimation (GAE) and Bias-Variance Trade-off

→ 1-Step TD Residual:

$$\delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$$

→ K-Step Advantage Estimation

$$\hat{A}_t^{(k)} = \sum_{l=0}^{k-1} \gamma^l \delta_{t+l}^V = -V(s_t) + \sum_{l=0}^{k-1} \gamma^l r_{t+l} + \gamma^k V(s_{t+k})$$

→ Generalized Advantage Estimation

$$\hat{A}_t^{GAE(\gamma,\lambda)} = (1-\lambda) \sum_{k=1}^{\infty} \lambda^{k-1} \hat{A}_t^{(k)} \; ; \quad \lambda \in [0,1]$$

(a.) Prove by algebraic substitution and grouping of powers γ and λ that:

$$\hat{A}_t^{GAE(\gamma,\lambda)} = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}^V$$

$$\hat{A}_t^{GAE(\gamma,\lambda)} = (1-\lambda)\left[\lambda^0 \hat{A}_t^{(1)} + \lambda^1 \hat{A}_t^{(2)} + \lambda^2 \hat{A}_t^{(3)} + \cdots\right]$$

$$= (1-\lambda)\left[\lambda^0(\gamma^0 \delta_{t+0}^V) + \lambda^1(\gamma^0 \delta_{t+0}^V + \gamma^1 \delta_{t+1}^V) + \lambda^2(\gamma^0 \delta_{t+0}^V + \gamma^1 \delta_{t+1}^V + \gamma^2 \delta_{t+2}^V) + \cdots\right]$$

$$= (1-\lambda)\left[\gamma^0 \delta_{t+0}^V(\lambda^0 + \lambda^1 + \lambda^2 + \cdots) + \lambda^1 \gamma^1 \delta_{t+1}^V(\lambda^0 + \lambda^1 + \cdots) + \lambda^2 \gamma^2 \delta_{t+2}^V(\lambda^0 + \lambda^1 + \lambda^2 + \cdots) + \cdots\right]$$

$$\hat{A}_t^{GAE(\gamma,\lambda)} = (1-\lambda)\left[\lambda^0\gamma^0\delta_{t+0}^V\sum_{m=0}^{\infty}\lambda^m + \lambda^1\gamma^1\delta_{t+1}^V\sum_{m=0}^{\infty}\lambda^m + \lambda^2\gamma^2\delta_{t+2}^V\sum_{m=0}^{\infty}\lambda^m + \cdots\right]$$

$$\sum_{m=0}^{\infty}\lambda^m = \frac{1}{1-\lambda} \;:\; |\lambda|<1$$

$$\therefore \hat{A}_t^{GAE(\gamma,\lambda)} = \left(\lambda^0\gamma^0\cdot\delta_{t+0}^V\cdot\frac{1}{1-\lambda} + \lambda^1\gamma^1\cdot\delta_{t+1}^V\cdot\frac{1}{1-\lambda} + \lambda^2\gamma^2\cdot\delta_{t+2}^V\cdot\frac{1}{1-\lambda} + \cdots\right)(1-\lambda)$$

$$\boxed{\hat{A}_t^{GAE(\gamma,\lambda)} = \sum_{l=0}^{\infty}(\lambda\gamma)^l\delta_{t+l}^V}$$

(b) Show $\hat{A}_t^{GAE(\lambda,\gamma)} = \delta_t^V$ when $\lambda = 0$

$$\hat{A}_t^{GAE(\gamma,\lambda=0)} = \hat{A}_t^{GAE(\gamma,0)}$$

$$= \sum_{l=0}^{\infty}\gamma^l\cdot(0)^l\cdot\delta_{t+l}^V$$

$$= \gamma^0\cdot(0)^0\cdot\delta_{t+0}^V + \underbrace{\sum_{l=1}^{\infty}\gamma^l\cdot(0)^l\cdot\delta_{t+l}^V}_{0}$$

$$= (1)\cdot(1)\cdot\delta_{t+0}^V$$

$$\therefore \boxed{\hat{A}_t^{GAE(\gamma,\lambda=0)} = \delta_t^V}$$

With $\lambda = 0$:

$$\hat{A}_t^{GAE(\gamma,0)} = \delta_t^V = r_t + \gamma V(s_{t+1}) - V(s_t)$$

$$\mathbb{E}_\pi\left[\hat{A}_t^{GAE(\gamma,0)} \mid s_t, a_t\right] = \mathbb{E}_\pi\left[r_t + \gamma V(s_{t+1}) - V(s_t) \mid s_t, a_t\right]$$

$$= \mathbb{E}_\pi\left[r_t + \gamma V(s_{t+1}) \mid s_t, a_t\right] - \mathbb{E}_\pi\left[V(s_t) \mid s_t, a_t\right]$$

$$= Q(s_t, a_t) - V(s_t)$$

True advantage is given by:

$$A^\pi(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t)$$

$\hat{A}_t^{GAE(\gamma,0)}$ is an unbiased estimator of the true advantage if and only if:

$$V^\pi(s_t) = V(s_t)$$
$$V^\pi(s_{t+1}) = V(s_{t+1})$$

Then

$$\mathbb{E}_\pi\left[\hat{A}_t^{GAE(\gamma,0)} \mid s_t, a_t\right] = A^\pi(s_t, a_t)$$

i.e. The data-roll out must be collected using an on-policy approach.

(c) when $\lambda = 1$

$$\hat{A}_t^{GAE(\gamma,\lambda=1)} = \hat{A}_t^{GAE(\gamma,1)} = \sum_{l=0}^{\infty} \gamma^l \delta_{t+l}^V$$

In infinite horizon, rollouts generated under policy π

$$\mathbb{E}_{\tau\sim\pi}\left[\hat{A}_t^{GAE(\gamma,1)} \mid s_t, a_t\right] = \mathbb{E}_{\tau\sim\pi}\left[\sum_{l=0}^{\infty} \gamma^l \delta_{t+l}^V \mid s_t, a_t\right]$$

$$= \mathbb{E}_{\tau\sim\pi}\left[\sum_{l=0}^{\infty} \gamma^l \left(\begin{array}{l} r_{t+l} \\ +\gamma V(s_{t+l+1}) \\ -V(s_{t+l}) \end{array}\right) \mid s_t, a_t\right]$$

$$= \mathbb{E}_{\tau\sim\pi}\left[\begin{array}{l} r_t + \gamma V(s_{t+1}) - V(s_t) \\ +\gamma r_{t+1} + \gamma^2 V(s_{t+2}) - \gamma V(s_{t+1}) \\ +\gamma^2 r_{t+2} + \gamma^3 V(s_{t+3}) - \gamma^2 V(s_{t+2}) \\ +\cdots \end{array} \mid s_t, a_t\right]$$

$$= \mathbb{E}_{\tau\sim\pi}\left[\begin{array}{l} r_t + \gamma r_{t+1} \\ +\gamma^2 r_{t+2} + \cdots \\ -V(s_t) \end{array} \mid s_t, a_t\right]$$

$$= \mathbb{E}_{\tau\sim\pi}\left[\left(\sum_{l=0}^{\infty} \gamma^l r_{t+l}\right) - V(s_t) \mid s_t, a_t\right]$$

$$= \mathbb{E}_{\tau\sim\pi}\left[\sum_{l=0}^{\infty} \gamma^l r_{t+l} \mid s_t, a_t\right] - \mathbb{E}_{\tau\sim\pi}\left[V(s_t) \mid s_t, a_t\right]$$

$$\mathbb{E}_{\tau\sim\pi}\left[\hat{A}_t^{GAE(\gamma,1)} \mid s_t, a_t\right] = Q^{\pi}(s_t,a_t) - V(s_t)$$

$$= \underbrace{Q^{\pi}(s_t,a_t) - V^{\pi}(s_t)} + V^{\pi}(s_t) - V(s_t)$$

$$\Rightarrow A^{\pi}(s_t,a_t)$$

$$\mathbb{E}_{\tau\sim\pi}\left[\hat{A}_t^{GAE(\gamma,1)} \mid s_t, a_t\right] = A^{\pi}(s_t,a_t) + V^{\pi}(s_t) - V(s_t)$$

To show the resulting policy gradient estimate is unbiased:

$$\mathbb{E}\left[\hat{A}_t^{GAE(\gamma,1)} \nabla_\theta \log \pi_\theta(a_t|s_t)\right]$$

$$= \mathbb{E}_{(s_t,a_t)}\left[\mathbb{E}\left[\hat{A}_t^{GAE(\gamma,1)} \nabla_\theta \log \pi_\theta(a_t|s_t) \mid s_t, a_t\right]\right]$$

$$= \mathbb{E}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t) \cdot \underbrace{\mathbb{E}\left[\hat{A}_t^{GAE(\gamma,1)} \mid s_t, a_t\right]}\right]$$

$$\Rightarrow A^{\pi}(s_t,a_t) + V^{\pi}(s_t) - V(s_t)$$

$$= \mathbb{E}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(s_t,a_t) A^{\pi}(s_t,a_t)\right]$$

$$+ \mathbb{E}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(s_t,a_t)\left(V^{\pi}(s_t) - V(s_t)\right)\right]$$

To be evaluated:

$$\mathbb{E}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(s_t,a_t)\left(V^{\pi}(s_t)-V(s_t)\right)\right] \to 0$$

must go to zero

Taking marginal over actions

$$\mathbb{E}_{s_t}\left[\mathbb{E}_{a_t\sim\pi(\cdot|s_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\left(V^{\pi}(s_t)-V(s_t)\,|\,s_t\right]\right]\right]$$

$$= \mathbb{E}_{s_t}\left[\left(V^{\pi}(s_t)-V(s_t)\right)\cdot \mathbb{E}_{a_t\sim\pi(\cdot|s_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\,|\,s_t\right]\right]$$

$$\nabla_\theta \log x(\theta) = \frac{1}{x(\theta)}\cdot \nabla_\theta x(\theta)$$

$$= \mathbb{E}_{s_t}\left[\left(V^{\pi}(s_t)-V(s_t)\right)\cdot \sum_{a_t} \pi_\theta(a_t|s_t)\cdot \frac{1}{\pi_\theta(a_t|s_t)}\cdot \nabla_\theta \pi_\theta(a_t|s_t)\right]$$

$$= \mathbb{E}_{s_t}\left[\left(V^{\pi}(s_t)-V(s_t)\right)\cdot \sum_{a_t} \nabla_\theta \pi_\theta(a_t|s_t)\right]$$

$$= \mathbb{E}_{s_t}\left[\left(V^{\pi}(s_t)-V(s_t)\right)\cdot \nabla_\theta \sum_{a_t} \pi_\theta(a_t|s_t)\right]$$

$$= \mathbb{E}_{s_t}\left[\left(V^{\pi}(s_t)-V(s_t)\right)\cdot \nabla_\theta(1)\right]$$

$$= \mathbb{E}_{s_t}\left[\left(V^{\pi}(s_t)-V(s_t)\right)\cdot 0\right] = 0$$

$$\therefore \nabla_\theta \mathbb{E}\left[\hat{A}_t^{GAE(\gamma,1)} \nabla_\theta \log \pi_\theta(a_t|s_t)\right]$$

$$= \mathbb{E}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t) A^{\pi}(a_t,s_t)\right]$$

$$+ \mathbb{E}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\left(V^{\pi}(s_t) - V(s_t)\right)\right]$$

$$= \mathbb{E}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t) \cdot A^{\pi}(a_t,s_t)\right]$$

Why does $\lambda=1$ lead to noisy policy-gradients:

With $\lambda=1$, the GAE advantage uses the full roll-out from monte-carlo. The full roll-out has higher variance as variance increases with the trajectory length or roll-out length. Then:

$$\hat{A}_t^{GAE(\gamma,1)} = \left(\sum_{l\geq 0}^{\infty} \gamma^l r_{t+l}\right) - V(s_t)$$

This removes all intermediate bootstrapped value predictions $V(s_{t+l})$ and the estimator depends on the complete sum of future empirical rewards over the entire episode trajectory.

Q2.] The clipped Surrogate Objective As A Pessimistic Lower Bound

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$$

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\left[\min\left(r_t(\theta)\hat{A}_t,\ clip(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right]$$

$$\epsilon \in [0, 1)$$

(a) Case 1: $\hat{A}_t > 0$; $r_t(\theta) > 1+\epsilon$

- $L^{CLIP}(\theta) = (1+\epsilon)\hat{A}_t$
- $\nabla_\theta L^{CLIP}(\theta) = \nabla_\theta\left((1+\epsilon)\hat{A}_t\right)$
  $= 0$

Case 2: $\hat{A}_t > 0$; $r_t(\theta) \leq 1+\epsilon$

- $L^{CLIP}(\theta) = r_t(\theta)\hat{A}_t$
- $\nabla_\theta L^{CLIP}(\theta) = \nabla_\theta\left(r_t(\theta)\hat{A}_t\right) = \hat{A}_t\,\nabla_\theta r_t(\theta)$
  $= \hat{A}_t \cdot \frac{\nabla_\theta \pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$

Case 3: $\hat{A}_t < 0$; $r_t(\theta) < 1-\epsilon$

- $L^{CLIP}(\theta) = (1-\epsilon)\hat{A}_t$
- $\nabla_\theta L^{CLIP}(\theta) = \nabla_\theta\left((1-\epsilon)\hat{A}_t\right)$
  $= 0$

Case 4: $\hat{A}_t < 0 \geq 1-\epsilon$

- $L^{CLIP}(\theta) = r_t(\theta)\hat{A}_t$
- $\nabla_\theta L^{CLIP}(\theta) = \nabla_\theta \left(r_t(\theta)\hat{A}_t\right)$
  $= \hat{A}_t \nabla_\theta r_t(\theta)$
  $= \hat{A}_t \nabla_\theta \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$

(b)

→ $\hat{A}_t > 0$

$clip(r_t(\theta), 1-\epsilon, 1+\epsilon) \leq 1+\epsilon$

$\therefore clip(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \leq (1+\epsilon)\hat{A}_t$

$\therefore \min(r_t(\theta)\hat{A}_t, clip(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t) \leq r_t(\theta)\hat{A}_t$

→ $\hat{A}_t < 0$

$clip(r_t(\theta), 1-\epsilon, 1+\epsilon) \geq 1-\epsilon$

$\therefore clip(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \leq (1-\epsilon)\hat{A}_t$

- If $r_t(\theta) \geq 1-\epsilon$

  $clip(r_t(\theta), 1-\epsilon, 1+\epsilon) \geq r_t(\theta)$

  $\therefore \min(r_t(\theta)\hat{A}_t, r_t(\theta)\hat{A}_t) = r_t(\theta)\hat{A}_t \leq r_t(\theta)\hat{A}_t$

- If $r_t(\theta) < 1-\epsilon$

  $clip(r_t(\theta), 1-\epsilon, 1+\epsilon) = 1-\epsilon$

  $\therefore \min(r_t(\theta)\hat{A}_t, (1-\epsilon)\hat{A}_t) = (1-\epsilon)\hat{A}_t \leq r_t(\theta)\hat{A}_t$

$\therefore$ $L^{CLIP}(\theta) \leq r_t(\theta)\hat{A}_t \; ; \forall\, r_t(\theta) > 0$ and $\hat{A}_t \in \mathbb{R}$

The $L^{CLIP}(\theta)$ acts as a pessimistic lower bound due to the pointwise minimum between the unclipped objective and the clipped objective.

For good actions ($\hat{A}_t > 0$), the $L^{CLIP}(\theta)$ caps the reward at $(1+\epsilon)\hat{A}_t$ and its min operator chooses this lower bound preventing inflation of the unclipped objective

For bad actions ($\hat{A}_t < 0$), the $L^{CLIP}(\theta)$ and its min operator select the smallest of $r_t(\theta)\hat{A}_t$ and $(1-\epsilon)\hat{A}_t$ allowing for policy update by selecting the more negative value of the two.

2. CONTINUOUS CONTROL IMPLEMENTATION IN MUJOCO

[0.5]

b) GAE Trade-off Ablation

Figure 2 shows a clear trade-off in convergence based on the value of λ. The default value of $\lambda = 0.95$ converges fastest to the maximum return of 1000 and is stable at that point.

The 1-step TD baseline ($\lambda = 0.0$) converges slowly, reaching the maximum return much later than for $\lambda = 0.95$. The evaluation for this 1-step TD baseline is relatively stable.

The full monte-carlo rollout ($\lambda=1.0$) is the slowest converging much more later than the 1-step TD baseline ($\lambda=0.0$) and the default value ($\lambda=0.95$). This monte-carlo estimation also shows high variance in its evolution before it finally converges.

These results confirm that $\lambda=0.95$ provides the best balance. It reduces variance relative to monte carlo while retaining enough information across steps to avoid bias that slows down convergence.

c) Policy Clipping Ablation

Disabling clipping as shown in figure 3 produces instability throughout the training process. In contrast, the clipped policy converges smoothly to 1000 and remains stable.

In figure 4, the KL-divergence plots makes the failure mode clear. The unclipped policy has large KL values compared to that of the clipped policy which stays at zero. This shows that without clipping, the optimizer takes excessively large policy update steps, the new policy drifts far from the old one in a single gradient step destroying previously learned behavior.

In continuous action spaces, the Gaussian policy's mean and variance can change significantly in a single update, causing the agent to suddenly explore regions of action space it had learned to avoid. The clipped surrogate objective prevents this by zeroing out gradient whenever the probability ratio is outside $[1-\varepsilon, 1+\varepsilon]$, effectively enforcing a trust region without requiring a constraint solver.

2 Continuous Control Implementation in MuJoCo

Q6 — Empirical Evaluation and Ablation Study

All experiments were run on the `InvertedPendulum-v5` MuJoCo environment for 1.2×10^5 environment interaction steps using the default hyperparameters ($\gamma = 0.99$, $\epsilon = 0.2$, $\lambda = 0.95$) unless otherwise stated.

2.1 Q6(a) — Benchmark: Default Hyperparameters (3 Seeds)

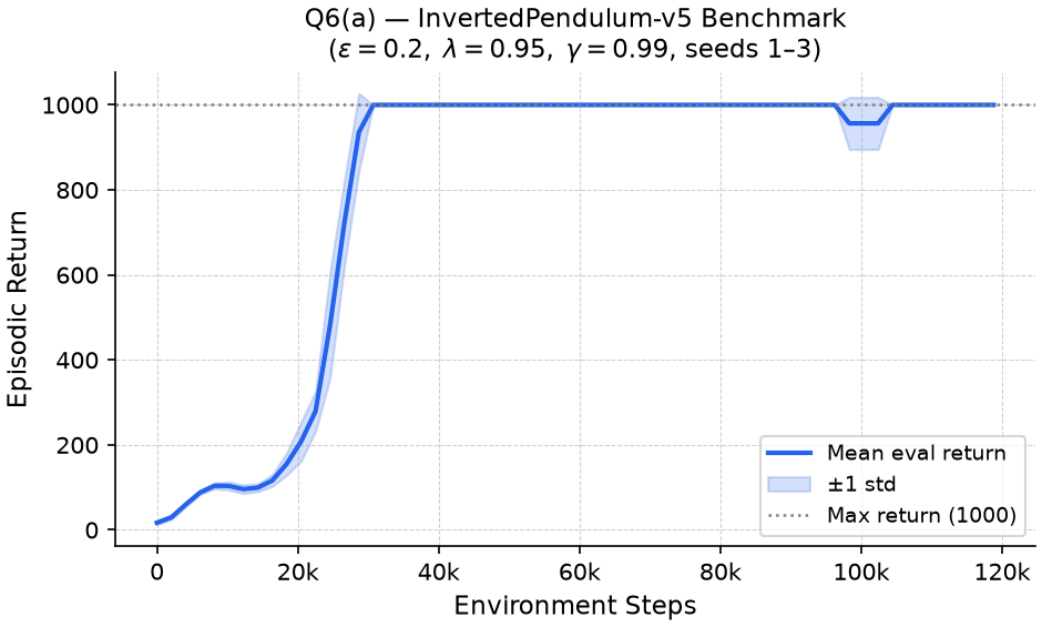


Figure 1: Evaluation episodic return (mean ± one standard deviation) over three random seeds (1, 2, 3) with default hyperparameters $\epsilon = 0.2$, $\lambda = 0.95$, $\gamma = 0.99$. The policy successfully balances the pendulum at the maximum return of 1000.

Figure 1 shows that the PPO implementation successfully learns to balance the inverted pendulum across all three seeds. The policy converges to the maximum return of 1000 by approximately 30k environment steps, with very low variance across seeds, demonstrating reliable and consistent training. A brief transient dip around 100k steps (visible in the shaded region) is a minor instability that resolves immediately, confirming the robustness of the clipped surrogate objective.

2.2 Q6(b) — GAE λ Trade-off Ablation

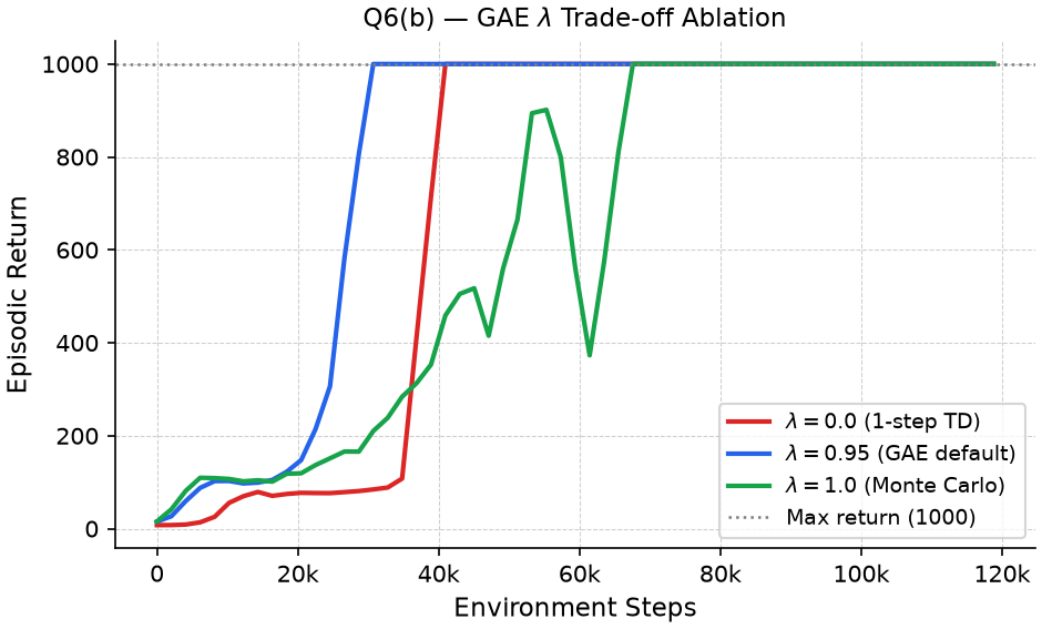


Figure 2: Learning curves for $\lambda \in \{0.0, 0.95, 1.0\}$ on `InvertedPendulum-v5`. All other hyperparameters are fixed at their default values.

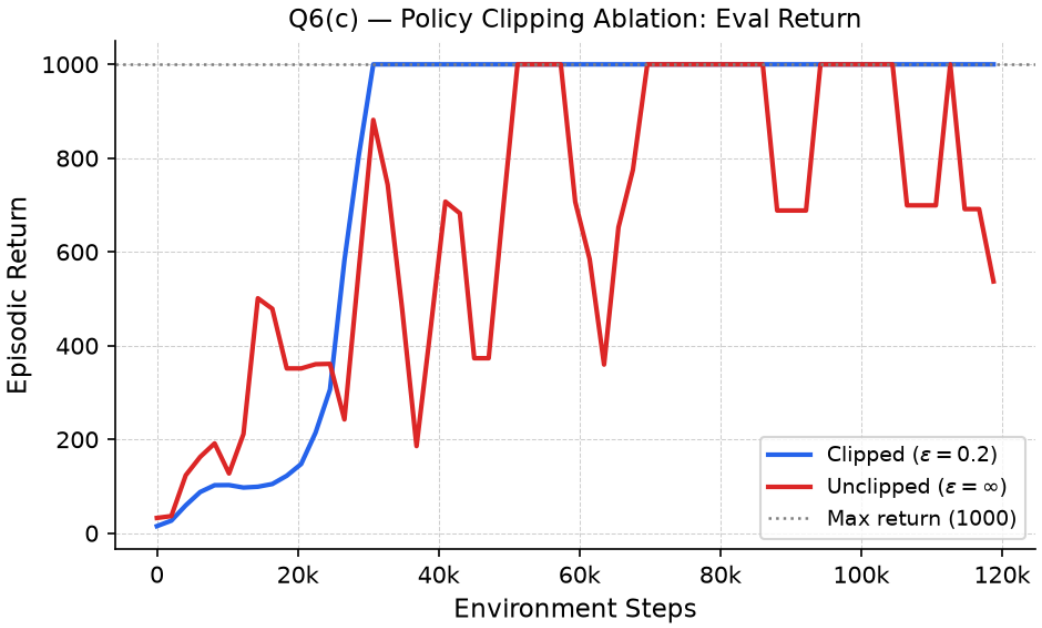
Figure 2 reveals a clear bias-variance trade-off governed by λ.

Convergence speed. The GAE default ($\lambda = 0.95$) converges fastest, reaching the maximum return of 1000 by ≈30k steps. The 1-step TD baseline ($\lambda = 0.0$) converges more slowly, not reaching maximum return until ≈40k steps. This is consistent with the high bias introduced by bootstrapping entirely from the value function: advantage estimates are myopic and less informative, slowing credit assignment across timesteps. The Monte Carlo estimator ($\lambda = 1.0$) is the slowest overall, not converging reliably until ≈80k steps.

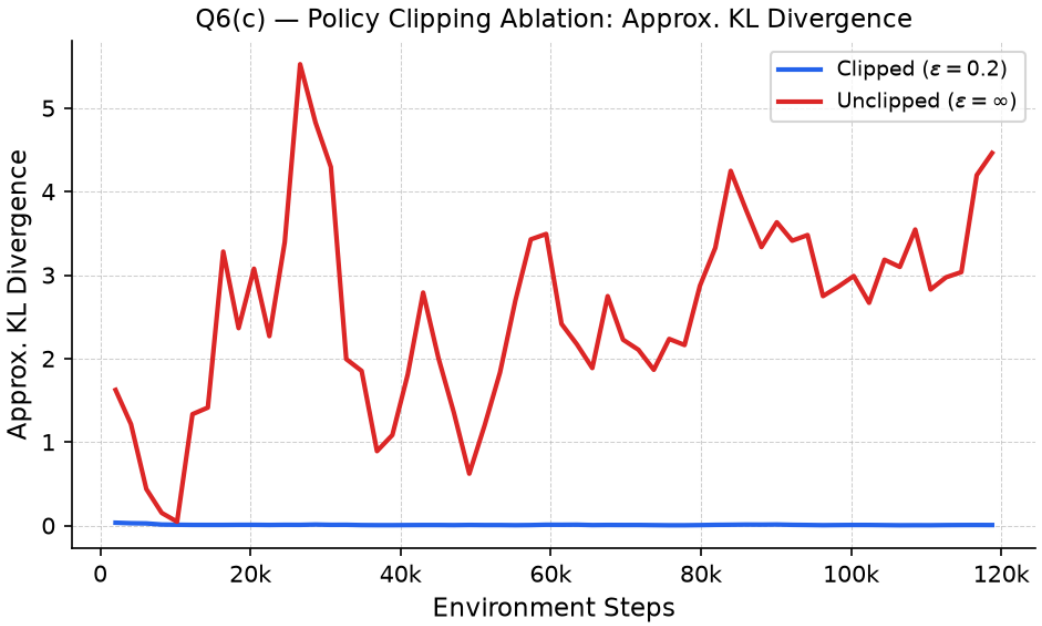
Variance. The Monte Carlo estimator ($\lambda = 1.0$) exhibits the highest variance by a large margin, with the learning curve oscillating dramatically between ≈400 and 900 before eventually stabilising. With $\lambda = 1.0$, all intermediate bootstrapped value predictions $V(s_{t+l})$ are removed and the estimator depends on the complete sum of future empirical rewards over the entire episode trajectory, whose variance increases with trajectory length. The 1-step TD baseline ($\lambda = 0.0$) is comparatively stable, as it bootstraps after a single step.

Final performance. All three variants reach the maximum return of 1000 asymptotically, so final performance is similar; the key difference lies in convergence speed and training stability. These results confirm that $\lambda = 0.95$ strikes the best empirical balance: it reduces variance relative to Monte Carlo while retaining enough multi-step return information to avoid the bias-induced slowdown of pure 1-step TD.

2.3 Q6(c) — Policy Clipping Ablation ($\epsilon = \infty$)



(a) Evaluation episodic return for clipped vs. unclipped policy.



(b) Approximate KL divergence between successive policies.

Figure 3: Policy clipping ablation on `InvertedPendulum-v5`. Clipped: $\epsilon = 0.2$; Unclipped: $\epsilon = 1000$ (effectively $\epsilon = \infty$). All other hyperparameters are fixed at their default values.

Disabling clipping ($\epsilon = \infty$) produces severe instability that persists throughout training. As shown in Figure 3a, the unclipped policy repeatedly reaches a return of 1000 but then

catastrophically collapses — dropping as low as 200–600 — before recovering, a cycle that continues even at 120k steps. In contrast, the clipped policy ($\epsilon = 0.2$) converges smoothly to 1000 by ≈30k steps and remains stable thereafter.

Failure mode. Figure 3b makes the failure mode explicit: the unclipped policy sustains approximate KL divergence values of 1–5 throughout training, with sharp spikes reaching ≈5.5, while the clipped policy's KL stays near 0 at all times. Without clipping, the optimizer takes excessively large policy update steps — the new policy drifts far from the old one within a single gradient step, destroying previously learned behaviour. In continuous action spaces this is especially damaging because the Gaussian policy's mean and variance can shift dramatically in a single update, causing the agent to explore regions of action space it had already learned to avoid. The clipped surrogate objective prevents this by zeroing out the gradient whenever the probability ratio moves outside $[1 - \epsilon, 1 + \epsilon]$, effectively enforcing a trust region without requiring a constraint solver.